

1) Vis at A er invertibel
og bestem A^{-1}

$$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{pmatrix}$$

"et Huk" $A \sim I \mid A^{-1}$

$$\begin{aligned} & \left(\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & -3 & 4 & 0 & -4 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 0 & 1 & 3 & -4 & 1 \end{array} \right) \\ & \sim \left(\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 0 & 1 & 3 & -4 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & -4 & 1 \end{array} \right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 4 & -1 \\ 0 & 1 & 0 & -2 & 5 & -2 \\ 0 & 0 & 1 & 3 & -4 & 1 \end{array} \right) \\ & \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 4 & -1 \\ 0 & 1 & 0 & -2 & 5 & -2 \\ 0 & 0 & 1 & 3 & -4 & 1 \end{array} \right) = I \mid A^{-1} \end{aligned}$$

2) $\text{las } Ax = b$

$b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ "sangeinverseri"

$A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 4 & -3 & 8 \end{pmatrix}$

for inverse of a her "3, 1" matrix $A^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 4 & -8 & 1 \\ -3 & 4 & -1 \end{pmatrix}$

$A^{-1}Ax = A^{-1}b$

$x = Ix = \begin{pmatrix} 1 & -1 & 1 \\ 4 & -8 & 1 \\ -3 & 4 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$

3) $\text{las } Ax = 2b$

$x = A^{-1}2b = \begin{pmatrix} 1 & -1 & 1 \\ 4 & -8 & 1 \\ -3 & 4 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix}$

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